

CASE STUDIES

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Section:1[Practical]

Task 1: Create and Manage Files

Objective: Practice creating, listing, viewing, and deleting files.

```
vaishnavi@Admin:~$ mkdir linux_task1
vaishnavi@Admin:~$ cd linux_task1
vaishnavi@Admin:~/linux_task1$ pwd
/home/vaishnavi/linux_task1
vaishnavi@Admin:~/linux_task1$ touch file1.txt file2.txt file3.txt file4.txt file5.txt
vaishnavi@Admin:~/linux_task1$ ls
file1.txt file2.txt file3.txt file4.txt file5.txt
vaishnavi@Admin:~/linux_task1$ ls -l
total 0
-rw-r--r-- 1 vaishnavi vaishnavi 0 Aug 12 04:56 file1.txt
-rw-r--r-- 1 vaishnavi vaishnavi 0 Aug 12 04:56 file2.txt
-rw-r--r-- 1 vaishnavi vaishnavi 0 Aug 12 04:56 file3.txt
-rw-r--r-- 1 vaishnavi vaishnavi 0 Aug 12 04:56 file4.txt
-rw-r--r-- 1 vaishnavi vaishnavi 0 Aug 12 04:56 file5.txt
vaishnavi@Admin:~/linux_task1$ echo "This is my first Linux file." > file1.txt
vaishnavi@Admin:~/linux_task1$ cat file1.txt
This is my first Linux file.
vaishnavi@Admin:~/linux_task1$ rm file5.txt
vaishnavi@Admin:~/linux_task1$ ls
file1.txt file2.txt file3.txt file4.txt
vaishnavi@Admin:~/linux_task1$ |
```

Task 2: Copy and Move Files

Objective: Practice copying and moving files and directories.

```

vaishnavi@Admin:~/linux_task2_new$ mkdir backup
vaishnavi@Admin:~/linux_task2_new$ ls
backup marks.txt notes.txt report.txt
vaishnavi@Admin:~/linux_task2_new$ cp report.txt backup/
vaishnavi@Admin:~/linux_task2_new$ ls backup
report.txt
vaishnavi@Admin:~/linux_task2_new$ cp marks.txt backup/marks_backup.txt
vaishnavi@Admin:~/linux_task2_new$ ls backup
marks_backup.txt report.txt
vaishnavi@Admin:~/linux_task2_new$ mv notes.txt backup/
vaishnavi@Admin:~/linux_task2_new$ mv report.txt final_report.txt
vaishnavi@Admin:~/linux_task2_new$ ls
backup final_report.txt marks.txt
vaishnavi@Admin:~/linux_task2_new$ ls -R
.:
backup final_report.txt marks.txt

./backup:
marks_backup.txt notes.txt report.txt
vaishnavi@Admin:~/linux_task2_new$ tree
.
├── backup
│   ├── marks_backup.txt
│   ├── notes.txt
│   └── report.txt
├── final_report.txt
└── marks.txt

2 directories, 5 files
vaishnavi@Admin:~/linux_task2_new$ |

```

Task 3: Working with head and tail

Objective: Practice viewing specific portions of a file

```
vaishnavi@Admin:~$ mkdir linux_task3
vaishnavi@Admin:~$ cd linux_task3
vaishnavi@Admin:~/linux_task3$ touch students.txt
vaishnavi@Admin:~/linux_task3$ ls
students.txt
vaishnavi@Admin:~/linux_task3$ nano students.txt
vaishnavi@Admin:~/linux_task3$ cat students.txt
Aarav
Ananya
Arjun
Diya
Haasini
Ishaan
Kavya
Manasa
Rahasya
Rahul
Sneha
Varun
Pranitha
Priya
Rohini
vaishnavi@Admin:~/linux_task3$ head -n 5 students.txt
Aarav
Ananya
Arjun
Diya
Haasini
vaishnavi@Admin:~/linux_task3$ head -n 10 students.txt
Aarav
Ananya
Arjun
Diya
```

```
Arjun
Diya
Haasini
vaishnavi@Admin:~/linux_task3$ head -n 10 students.txt
Aarav
Ananya
Arjun
Diya
Haasini
Ishaan
Kavya
Manasa
Rahasya
Rahul
vaishnavi@Admin:~/linux_task3$ tail -n 5 students.txt
Sneha
Varun
Pranitha
Priya
Rohini
vaishnavi@Admin:~/linux_task3$ tail -n 10 students.txt
Ishaan
Kavya
Manasa
Rahasya
Rahul
Sneha
Varun
Pranitha
Priya
Rohini
vaishnavi@Admin:~/linux_task3$ |
```

Task 4: Organize Files

Objective: Organize files into different directories using mv

```

vaishnavi@Admin:~$ mkdir organization
vaishnavi@Admin:~$ cd organization
vaishnavi@Admin:~/organization$ touch file1.txt file2.txt file3.txt image1.jpg image2.jpg
vaishnavi@Admin:~/organization$ ls
file1.txt file2.txt file3.txt image1.jpg image2.jpg
vaishnavi@Admin:~/organization$ mkdir documents
vaishnavi@Admin:~/organization$ mkdir images
vaishnavi@Admin:~/organization$ mkdir images
mkdir: images: File exists
vaishnavi@Admin:~/organization$ mv *.txt documents/
vaishnavi@Admin:~/organization$ ls documents
file1.txt file2.txt file3.txt
vaishnavi@Admin:~/organization$ mv *.jpg images/
vaishnavi@Admin:~/organization$ ls images
image1.jpg image2.jpg
vaishnavi@Admin:~/organization$ ls documents
file1.txt file2.txt file3.txt
vaishnavi@Admin:~/organization$ ls images
image1.jpg image2.jpg
vaishnavi@Admin:~/organization$ mv documents/file1.txt documents/notes.txt
vaishnavi@Admin:~/organization$ mv images/image1.jpg images/photo.jpg
vaishnavi@Admin:~/organization$ mv images/image1.jpg images/photo.jpg
mv: cannot stat 'images/image1.jpg': No such file or directory
vaishnavi@Admin:~/organization$ tree
.
├── documents
│   ├── file2.txt
│   ├── file3.txt
│   └── notes.txt
└── images
    ├── image2.jpg
    └── photo.jpg

3 directories, 5 files
vaishnavi@Admin:~/organization$ |

```

Task 5: Directory Management

Objective: Practice creating nested directories and removing them

```
pranitha@Pranitha:~$ mkdir project
pranitha@Pranitha:~$ cd project
pranitha@Pranitha:~/project$ mkdir -p source/c source/python docs backup
pranitha@Pranitha:~/project$ ls
backup docs source
pranitha@Pranitha:~/project$ ls -R
.:
backup docs source

./backup:

./docs:

./source:
c python

./source/c:

./source/python:
pranitha@Pranitha:~/project$ touch source/c/file1.c source/c/file2.c source/c/file3.c
pranitha@Pranitha:~/project$ ls source/c
file1.c file2.c file3.c
pranitha@Pranitha:~/project$ touch source/python/file1.py source/python/file2.py
pranitha@Pranitha:~/project$ ls source/python
file1.py file2.py
pranitha@Pranitha:~/project$ ls -R
.:
backup docs source

./backup:

./docs:

./source:
c python

./source/c:
file1.c file2.c file3.c

./source/python:
file1.py file2.py
```

```

vaishnavi@Admin:~$ mkdir project
vaishnavi@Admin:~$ cd project
vaishnavi@Admin:~/project$ mkdir source
vaishnavi@Admin:~/project$ mkdir docs
vaishnavi@Admin:~/project$ mkdir backup
vaishnavi@Admin:~/project$ mkdir source/c
vaishnavi@Admin:~/project$ mkdir source/python
vaishnavi@Admin:~/project$ touch source/c/file1.c source/c/file2.c source/c/file3.c
vaishnavi@Admin:~/project$ touch source/python/file1.py source/python/file2.py
vaishnavi@Admin:~/project$ tree
.
├── backup
├── docs
└── source
    ├── c
    │   ├── file1.c
    │   ├── file2.c
    │   └── file3.c
    └── python
        ├── file1.py
        └── file2.py

6 directories, 5 files
vaishnavi@Admin:~/project$ rmdir backup
vaishnavi@Admin:~/project$ mkdir backup
vaishnavi@Admin:~/project$ touch backup/test.txt
vaishnavi@Admin:~/project$ ls backup
test.txt
vaishnavi@Admin:~/project$ rmdir backup
rmdir: failed to remove 'backup': Directory not empty
vaishnavi@Admin:~/project$ rm backup/test.txt
vaishnavi@Admin:~/project$ rmdir backup
vaishnavi@Admin:~/project$ tree
.
├── docs
└── source
    ├── c
    │   ├── file1.c
    │   ├── file2.c
    │   └── file3.c
    └── python
        ├── file1.py
        └── file2.py

5 directories, 5 files
vaishnavi@Admin:~/project$ |

```

Task 6: File Permissions Using chmod

Objective: Understand Linux file permissions.

```

vaishnavi@Admin:~$ cd ~
vaishnavi@Admin:~$ mkdir linux_task4
mkdir: linux_task4: File exists
vaishnavi@Admin:~$ cd linux_task4
vaishnavi@Admin:~/linux_task4$ touch public.txt private.txt script.sh
vaishnavi@Admin:~/linux_task4$ ls
private.txt  public.txt  script.sh
vaishnavi@Admin:~/linux_task4$ echo "This is a public file." > public.txt
vaishnavi@Admin:~/linux_task4$ echo "This is a private file." > private.txt
vaishnavi@Admin:~/linux_task4$ echo "echo Hello from script" > script.sh
vaishnavi@Admin:~/linux_task4$ ls -l
total 12
-rw----- 1 vaishnavi vaishnavi 24 Aug 12 05:24 private.txt
-rw-r--r-- 1 vaishnavi vaishnavi 23 Aug 12 05:24 public.txt
-rw-r--r-- 1 vaishnavi vaishnavi 23 Aug 12 05:24 script.sh
vaishnavi@Admin:~/linux_task4$ chmod 600 private.txt
vaishnavi@Admin:~/linux_task4$ ls -l private.txt
-rw----- 1 vaishnavi vaishnavi 24 Aug 12 05:24 private.txt
vaishnavi@Admin:~/linux_task4$ chmod 644 public.txt
vaishnavi@Admin:~/linux_task4$ ls -l public.txt
-rw-r--r-- 1 vaishnavi vaishnavi 23 Aug 12 05:24 public.txt
vaishnavi@Admin:~/linux_task4$ chmod +x script.sh
vaishnavi@Admin:~/linux_task4$ ls -l
total 12
-rw----- 1 vaishnavi vaishnavi 24 Aug 12 05:24 private.txt
-rw-r--r-- 1 vaishnavi vaishnavi 23 Aug 12 05:24 public.txt
-rwxr-xr-x 1 vaishnavi vaishnavi 23 Aug 12 05:24 script.sh
vaishnavi@Admin:~/linux_task4$ chmod -x script.sh
vaishnavi@Admin:~/linux_task4$ ls -l script.sh
-rw-r--r-- 1 vaishnavi vaishnavi 23 Aug 12 05:24 script.sh
vaishnavi@Admin:~/linux_task4$ |

```

Task 7: File Permission Challenge

Objective: Understand how permissions affect files and directories.

```

vaishnavi@Admin:~$ mkdir permission_lab
vaishnavi@Admin:~$ cd permission_lab
vaishnavi@Admin:~/permission_lab$ touch test.txt
vaishnavi@Admin:~/permission_lab$ ls -l
total 0
-rw-r--r-- 1 vaishnavi vaishnavi 0 Aug 12 05:26 test.txt
vaishnavi@Admin:~/permission_lab$ echo "This is a permission test file." > test.txt
vaishnavi@Admin:~/permission_lab$ cat test.txt
This is a permission test file.
vaishnavi@Admin:~/permission_lab$ chmod u-r test.txt
vaishnavi@Admin:~/permission_lab$ ls -l test.txt
--w-r--r-- 1 vaishnavi vaishnavi 32 Aug 12 05:27 test.txt
vaishnavi@Admin:~/permission_lab$ cat test.txt
cat: test.txt: Permission denied
vaishnavi@Admin:~/permission_lab$ chmod u+r test.txt
vaishnavi@Admin:~/permission_lab$ cat test.txt
This is a permission test file.
vaishnavi@Admin:~/permission_lab$ chmod u-w test.txt
vaishnavi@Admin:~/permission_lab$ ls -l test.txt
-r--r--r-- 1 vaishnavi vaishnavi 32 Aug 12 05:27 test.txt
vaishnavi@Admin:~/permission_lab$ echo "New line" >> test.txt
-bash: test.txt: Permission denied
vaishnavi@Admin:~/permission_lab$ chmod u+w test.txt
vaishnavi@Admin:~/permission_lab$ echo "New line" >> test.txt
vaishnavi@Admin:~/permission_lab$ cat test.txt
This is a permission test file.
New line
vaishnavi@Admin:~/permission_lab$ chmod u-x test.txt
vaishnavi@Admin:~/permission_lab$ ls -l test.txt
-rw-r--r-- 1 vaishnavi vaishnavi 41 Aug 12 05:28 test.txt
vaishnavi@Admin:~/permission_lab$ ./test.txt
-bash: ./test.txt: Permission denied
vaishnavi@Admin:~/permission_lab$ |

```

Task 8: Backup and Restore

Objective: Create backups and restore files using cp

```

vaishnavi@Admin:~$ cd ~
vaishnavi@Admin:~$ mkdir linux_task6
vaishnavi@Admin:~$ cd linux_task6
vaishnavi@Admin:~/linux_task6$ touch important.txt
vaishnavi@Admin:~/linux_task6$ nano important.txt
vaishnavi@Admin:~/linux_task6$ mkdir backup
vaishnavi@Admin:~/linux_task6$ ls
backup  important.txt
vaishnavi@Admin:~/linux_task6$ cp important.txt backup/
vaishnavi@Admin:~/linux_task6$ ls backup
important.txt
vaishnavi@Admin:~/linux_task6$ mv backup/important.txt backup/important_backup.txt
vaishnavi@Admin:~/linux_task6$ ls backup
important_backup.txt
vaishnavi@Admin:~/linux_task6$ rm important.txt
vaishnavi@Admin:~/linux_task6$ ls
backup
vaishnavi@Admin:~/linux_task6$ cp backup/important_backup.txt .
vaishnavi@Admin:~/linux_task6$ ls
backup  important_backup.txt
vaishnavi@Admin:~/linux_task6$ mv important_backup.txt important_restored.txt
vaishnavi@Admin:~/linux_task6$ ls
backup  important_restored.txt
vaishnavi@Admin:~/linux_task6$ cat important_restored.txt
This is important information.
Linux is an operating system.
Files can be backed up.
Backup protects important data.
The cp command copies files.
The mv command moves files.
The rm command deletes files.
Restoring data is useful.
Linux file management is easy.
This is my backup practice.
vaishnavi@Admin:~/linux_task6$ |

```

Task 9: Complete Linux File Management Challenge

Objective: Use almost all the commands together.

```

vaishnavi@Admin:~$ mkdir linux_project
vaishnavi@Admin:~$ cd linux_project
vaishnavi@Admin:~/linux_project$ pwd
/home/vaishnavi/linux_project
vaishnavi@Admin:~/linux_project$ mkdir source backup docs
vaishnavi@Admin:~/linux_project$ ls
backup  docs  source
vaishnavi@Admin:~/linux_project$ touch source/main.c source/helper.c source/README.txt
vaishnavi@Admin:~/linux_project$ ls source
README.txt  helper.c  main.c
vaishnavi@Admin:~/linux_project$ nano source/README.txt
vaishnavi@Admin:~/linux_project$ head -n 5 source/README.txt
Linux is an operating system.
Linux supports file management.
Directories organize files.
The cp command copies files.
The mv command moves files.
vaishnavi@Admin:~/linux_project$ tail -n 5 source/README.txt
Linux is open source.
The terminal runs commands.
File management is important.
This is a practical task.
Linux practice is useful.
vaishnavi@Admin:~/linux_project$ cp source/main.c backup/
vaishnavi@Admin:~/linux_project$ ls backup
main.c
vaishnavi@Admin:~/linux_project$ cp source/README.txt docs/
vaishnavi@Admin:~/linux_project$ ls docs
README.txt
vaishnavi@Admin:~/linux_project$ mv source/helper.c source/functions.c
vaishnavi@Admin:~/linux_project$ ls source
README.txt  functions.c  main.c
vaishnavi@Admin:~/linux_project$ chmod 644 source/README.txt
vaishnavi@Admin:~/linux_project$ ls -l source/README.txt
-rw-r--r-- 1 vaishnavi vaishnavi 565 Aug 12 05:32 source/README.txt
vaishnavi@Admin:~/linux_project$ rm backup/main.c
vaishnavi@Admin:~/linux_project$ rmdir backup
vaishnavi@Admin:~/linux_project$ ls
docs  source
vaishnavi@Admin:~/linux_project$ ls -R
.:
docs  source

./docs:
README.txt

./source:
README.txt  functions.c  main.c
vaishnavi@Admin:~/linux_project$ tree
.
├── docs
│   └── README.txt
└── source
    ├── README.txt
    ├── functions.c
    └── main.c

3 directories, 4 files
vaishnavi@Admin:~/linux_project$ |

```

Task 10: Mini Linux Administration Challenge

Objective: Complete a real-world-style file management exercise.

```

vaishnavi@Admin:~$ mkdir student_portal
vaishnavi@Admin:~$ cd student_portal
vaishnavi@Admin:~/student_portal$ pwd
/home/vaishnavi/student_portal
vaishnavi@Admin:~/student_portal$ mkdir src docs backup logs config
vaishnavi@Admin:~/student_portal$ ls
backup  config  docs  logs  src
vaishnavi@Admin:~/student_portal$ touch src/main.c src/student.c docs/README.txt config/config.txt logs/system.log
vaishnavi@Admin:~/student_portal$ ls -R
.:
backup  config  docs  logs  src

./backup:

./config:
config.txt

./docs:
README.txt

./logs:
system.log

./src:
main.c  student.c
vaishnavi@Admin:~/student_portal$ nano logs/system.log
vaishnavi@Admin:~/student_portal$ tail -n 5 logs/system.log
User logged out successfully.
New student registered.
Records updated successfully.
System check completed.
Portal ready.
vaishnavi@Admin:~/student_portal$ cp docs/README.txt backup/
vaishnavi@Admin:~/student_portal$ ls backup
README.txt
vaishnavi@Admin:~/student_portal$ mv backup/README.txt backup/README_backup.txt
vaishnavi@Admin:~/student_portal$ chmod 600 config/config.txt
vaishnavi@Admin:~/student_portal$ ls -l config/config.txt
-rw----- 1 vaishnavi vaishnavi 0 Aug 12 05:35 config/config.txt
vaishnavi@Admin:~/student_portal$ chmod 644 docs/README.txt
vaishnavi@Admin:~/student_portal$ ls -l docs/README.txt
-rw-r--r-- 1 vaishnavi vaishnavi 0 Aug 12 05:35 docs/README.txt
vaishnavi@Admin:~/student_portal$ rm backup/README_backup.txt
vaishnavi@Admin:~/student_portal$ ls backup
vaishnavi@Admin:~/student_portal$ rmdir backup
vaishnavi@Admin:~/student_portal$ ls
config  docs  logs  src
vaishnavi@Admin:~/student_portal$ ls -R
.:
config  docs  logs  src

./config:
config.txt

./docs:
README.txt

./logs:
system.log

./src:
main.c  student.c
vaishnavi@Admin:~/student_portal$ tree
.
├── config
│   └── config.txt
├── docs
│   └── README.txt
├── logs
│   └── system.log
└── src
    ├── main.c
    └── student.c

5 directories, 5 files
vaishnavi@Admin:~/student_portal$ |

```